Colloids and Surfaces A: Physicochemical and Engineering Aspects 738 (2026) 139891

Contents lists available at ScienceDirect

Colloids and Surfaces A: Physicochemical and Engineering Aspects

journal homepage: www.elsevier.com/locate/colsurfa

Spontaneous surfactant solution imbibition kinetics in oil-saturated nanoporous media

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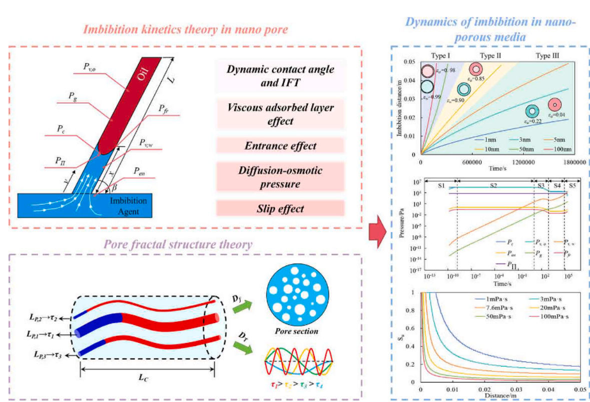
HIGHLIGHTS

- Kinetic model for surfactant solution imbibition in oil-saturated nanopore is established.
- IFT and diffusion-osmotic pressure are key factors for nanoscale imbibition.
- Imbibition process is divided into five stages based on force competition.
- Slip effect dominates imbibition under high-energy conditions.
- Diffusion-osmotic pressure drives imbibition under low-energy conditions.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Confinement effects
Imbibition kinetics
Nanoporous media
Fractal theory
Analytical model

ABSTRACT

Spontaneous imbibition is a fundamental process governing fluid flow in nanoporous media, where interfacial and confinement effects are dominant. However, existing models fail to adequately describe surfactant transport within nanopores. Integrating dynamic contact angle, interfacial tension (IFT), viscous adsorbed layer effect, slip effect, and diffusion-osmotic pressure, this study establishes a comprehensive kinetic model that bridges nanoscale flow mechanisms with core-scale fractal characteristics. Sensitivity analysis reveals that IFT and diffusion-osmotic pressure are the primary factors governing nanoscale imbibition, followed by contact angle, pore radius, and oil viscosity. The imbibition process is controlled by the competition between driving and resisting pressures and can be divided into five distinct stages. This reflects the process of the dominant mechanism evolving from inertia in the early stage to viscosity in the later stage, revealing the special mechanism of fluid flow within a confined nanopore. Application of fractal theory enables quantitative evaluation of core-scale imbibition efficiency and sweep distance. The results demonstrate that under high-energy conditions with sufficient imbibition driving pressure, the slip effect becomes the dominant mechanism. Conversely, under low-energy conditions, diffusion-osmotic pressure serves as a main driving mechanism and plays a crucial role. This research offers new insights and methods for understanding surfactant-driven imbibition dynamics in nanoporous media.

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https://doi.org/10.1016/j.colsurfa.2026.139891
Received 16 December 2025; Received in revised form 28 January 2026; Accepted 6 February 2026
Available online 18 February 2026

1. Introduction

Two-phase flow in porous media is a natural phenomenon that is widely present in carbon dioxide sequestration [1], groundwater seepage [2], crude oil recovery [3,4], and drug delivery processes [5]. Spontaneous imbibition, where a wetting phase displaces a non-wetting phase driven by capillary pressure, is governed by the interplay of capillary, viscous, and gravity. However, in nanoporous media, the coupling of confinement [6], entrance [7,8], dynamic wetting [9], and slip effect [10] renders the traditional Lucas-Washburn model inapplicable, especially with the addition of surfactants.

Research in this field primarily relies on experiments [11], simulations [12,13], and analytical modeling [14]. While experiments offer objective insights, they are often constrained by high complexity and limited reproducibility [15]. Similarly, molecular simulations are computationally intensive, restricting their application at larger scales [16]. Consequently, analytical modeling has emerged as an efficient tool for elucidating nanoscale imbibition mechanisms [17].

In the early 20th century, Lucas, Washburn, and Ridal established an idealized physical model (LWR model) for the spontaneous imbibition process in complex porous media by balancing capillary pressure and viscous pressure drop in capillaries, and first revealed the relationship that the imbibition distance is proportional to the square root of time [18,19]. Chibbaro [20] et al. demonstrated that the smaller the pore/capillary radius, the greater the deviation between the classical theory and the experiment, indicating that the classical model fails to describe imbibition in nanopores. Subsequent studies introduced various correction terms to the LWR model to better represent nanoscale imbibition. Li [21] et al. believe that due to the influence of shear stress and frictional resistance during the imbibition process, the contact angle should be dynamic, and thus introduced dynamic wettability into the imbibition model. Heiranian [22] highlighted that when the slip length in nanopores is significant, the entrance effect also becomes non-negligible and modified the traditional L-W model for carbon nanotubes accordingly. Wang [23] et al. proposed a new solution for spontaneous imbibition considering dynamic contact angle and gravity, and extended it to core-scale analysis using the fractal dimension of pore radius. Zhao [24] et al. developed a fractal porous model emphasizing the role of osmotic pressure in clay minerals during imbibition. Yang [25] et al. established a fractal model of fluid imbibition into irregular rough pores without gravity and under gravity, and derived expressions for imbibition height and imbibition mass in irregular rough pores. Subsequently, researchers conducted a sensitivity analysis to systematically evaluate the influence of relative roughness, gravity, fractal dimension, and shape factor on the imbibition dynamics. Dong [26] et al. derived a comprehensive spontaneous imbibition model at the fracture and core scales considering rough wall geometry, streamline tortuosity, and heterogeneity of the fracture size distribution by applying the classical cubic law describing flat flow and fractal geometry theory. Tian [27] et al. further incorporated the effects of dynamic contact angle, nanoconfinement, inertia, and entrance conditions, and used a lognormal distribution to characterize pore structures in core-scale imbibition. This model is among the most comprehensive currently available; however, a fully unified model that integrates all these effects remains lacking.

Surfactants are amphiphilic organic molecules containing both hydrophilic and hydrophobic groups. Depending on the type of hydrophilic group, they are typically classified as anionic, cationic, or nonionic surfactants [28,29]. By adsorbing at the oil-water interface and rock surface, surfactants reduce interfacial tension (IFT) and alter wettability, thereby enhancing imbibition efficiency and sweep performance [11,30,31]. Furthermore, the IFT at the oil-water interface evolves dynamically over time [32,33], and ionic surfactants can generate additional diffusion-osmotic pressure at the interface [34,35]. These phenomena are also significant in nanopores. The pore size distribution of natural cores shows obvious multi-scale characteristics, and the flow channels have significant tortuosity in space. Moreover, the pore radius also affects tortuosity; generally, the smaller the pore radius, the greater the tortuosity [36,37]. Therefore, integrating the single-capillary imbibition model into a more universal core-scale framework remains a key research challenge.

To overcome this limitation, this research develops a multi-scale analytical framework that systematically couples the nanoscale interfacial dynamics with the structural heterogeneity of nanoporous media. In this framework, a dynamic model was developed by incorporating inertia, dynamic contact angle/IFT, viscous adsorbed layer effect, slip effect, entrance effect modified by slip effect, and diffusion-osmotic pressure. Based on fractal theory, an analytical mesoscale model suitable for nanoporous media was further constructed by accounting for the distributions of pore radius and tortuosity and integrating the nanoscale imbibition model. The effects of pore size, IFT, and oil-phase viscosity on imbibition distance and efficiency were systematically analyzed, and the underlying mechanical mechanisms were elucidated.

2. Establishment of imbibition theoretical model

The spontaneous imbibition process in deep reservoirs typically occurs under constant temperature conditions, and fluid compressibility is negligible compared to the strong capillary driving forces in nanopores [38]. Moreover, natural sedimentary rocks exhibit distinct self-similar pore structures. Fractal theory provides an effective mathematical tool to quantitatively characterize the multi-scale distribution and complex geometry of these pores, allowing for the upscaling of single-pore laws to the core scale [39]. So to describe the oil-water flow process within the nanopores, the following assumptions are made:

(1) The flow is isothermal and incompressible, and the effects of oil and water evaporation are neglected. The flow follows Poiseuille's law.

(2) The rock core pores are simplified as a network of capillary channels with varying radii and tortuosities. These capillaries are randomly oriented within the rock core. The fractal dimension is used to characterize the pore size distribution and spatial filling features, while tortuosity represents the non-linear geometry of the channels.

Based on the above assumptions, this research first establishes a single capillary imbibition model and then extends it to a fractal capillary network to obtain the overall imbibition volume of the rock core.

2.1. Derivation of control equation for nanopore imbibition process

The tight reservoirs are mainly composed of nano-pores, where a series of micro-scale flow effects and nano-confinement effects exist. The addition of surfactants significantly affects the oil-water interface properties, and the traditional LWR equation is difficult to describe the oil-water displacement process in nanopores. This section mainly considers the slip effect, the slip effect-corrected entrance effect, viscous adsorbed layer effect, the dynamic contact angle/IFT, and the diffusion-osmotic pressure of the surfactant solution to establish a mechanism model for surfactant imbibition in nanopores.Fig. 1.

Then, the imbibition process should consider the inertial pressure P_{in}, capillary pressure P_c, viscous pressure drop of the oil phase $P_{v,\ o}$, viscous pressure drop of the water phase $P_{v,\ w}$, resistance P_{en} generated by the entrance effect, gravity P_g, friction pressure P_{fr} of the three-phase contact line, and diffusion-osmotic pressure $P_{\prod}$. The model established in this paper can be expressed as:

$$p_{in} = p_c - p_{v,o} - p_{v,w} - p_{en} - p_g - p_{fr} + p_{\prod} \tag{1}$$

2.1.1. Inertial pressure

The inertial resistance generated by Newton's second law when the fluid accelerates/decelerates. The inertial pressure mainly plays a major role in the early stage of imbibition, when the migration speed of the oil-water front gradually increases and when the interface has not yet been significantly slowed down by viscous resistance. The inertial pressure

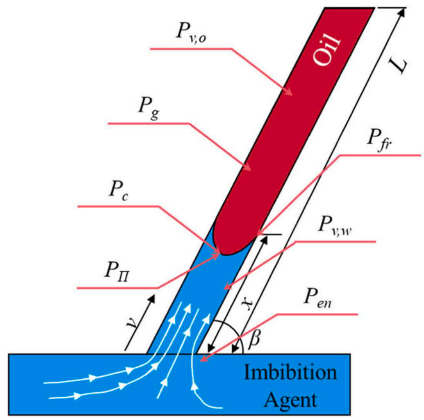

Fig. 1. Controlled pressure of spontaneous surfactant solution imbibition in oil-saturated nanopore.

can be expressed as [27]

$$p_{in} = \frac{d[\rho_w x v + \rho_o (L - x) v]}{dt} \tag{2}$$

Where, ρ_w is the water phase density, kg/m^3; ρ_o is the oil phase density, kg/m^3; x is the imbibition distance, m; v is the instantaneous moving velocity of the wetting front (i.e., the water-oil interface), m/s; L is the pore length, m.

2.1.2. Viscous pressure drop generated by oil/water phase flow

When oil/water two-phase flows in porous media or capillaries, the internal friction between liquid molecules generates viscous pressure drop, which manifests as a pressure drop along the flow path. Considering the slip effect in nanopores, the fluid flow velocity changes near the boundary. According to the modified Poiseuille's law [40], the volumetric flow rate under slip conditions, denoted as Q_{SLIP}, is expressed as:

$$Q_{\mathrm{SLIP}} = \frac{\pi[r^4 + 4r^3 L_S]}{8\mu}\frac{p}{L} = \frac{\pi r^4}{8\mu L}\left(1 + \frac{4L_s}{r}\right)p \tag{3}$$

Where, Q_{SLIP} represents the fluid flow rate under slip conditions, m^3/s; r represents the pore radius, m; L_S represents the slip distance, m; μ represents the fluid viscosity, Pa·s; P represents the flow pressure difference, Pa.

The viscous pressure drop caused by the flow of water is

$$p_{v,w} = \frac{Q}{\frac{\pi r^4}{8\mu_{\mathrm{eff,w}} x}\left(1 + \frac{4L_{s,w}}{r}\right)} = \frac{8\mu_{\mathrm{eff,w}} x}{(r^2 + 4L_{s,w} r)}\mathrm{v} \tag{4}$$

Where, $\mu_{eff,w}$ represents the effective viscosity of the water phase, Pa·s; $L_{s,w}$ represents the slip distance of the water phase, m.

Huang proposed the relationship between the water phase slip distance and the contact angle.

$$L_{s,w} = \frac{\mathrm{C}_1}{(\cos\theta + 1)^2} \tag{5}$$

Where, C_1 is a dimensionless constant determined by molecular dynamics simulation to be 0.41 [41,42]; θ is the contact angle, °.

Similarly, the viscous pressure drop caused by oil phase flow is

$$p_{v,o} = \frac{Q}{\frac{\pi r^4}{8\mu_{\mathrm{eff,o}}(L-x)}\left(1 + \frac{4L_{s,o}}{r}\right)} = \frac{8\mu_{\mathrm{eff,o}}(L - x)}{(r^2 + 4L_{s,o} r)}\mathrm{v} \tag{6}$$

Where, $\mu_{eff,\ o}$ represents the effective viscosity of the water phase, Pa·s; $L_{s,\ o}$ represents the slip distance of the oil phase, m.

McBride et al. proposed an empirical relationship between the slip distance and viscosity of alkanes [43].

$$\mathrm{L}_{s,o} = A\mu^x \tag{7}$$

Where, A, x represents the fitting coefficient, dimensionless, where A can be 130 and x can be 0.33.

Raviv et al. proposed a fitting relationship between bulk viscosity, surface viscosity and contact angle [44].

$$\frac{\mu_i}{\mu_B} = -0.018\theta + 3.25 \tag{8}$$

Where, μ_i represents the surface viscosity, Pa·s; μ_B represents the bulk viscosity, Pa·s.

The effective viscosity is calculated using the area weighted method, where the thickness of the boundary fluid is the thickness of two molecular effective diameters [45] :

$$\mu_{\mathrm{eff}} = \mu_{\mathrm{i}}\frac{A_i}{A_t} + \mu_B\left(1 - \frac{A_i}{A_t}\right) \tag{9}$$

Where, A_t represents the area of the bulk region, m^3, $A_t = \pi r^2$; A_i represents the area of the interface region, m^3, $A_i = \pi r^2 - \pi(r-r_c)^2$.

This paper uses the Chapman-Enskog collision diameter to represent the effective diameter of a molecule. According to the Lennard-Jones potential theory, the effective diameter of a water molecule is $\sigma_{CE,\ water}$= 2.641 Å [46]. Messerly et al. calculated the Chapman-Enskog collision diameters of CH_3 and CH_2 chains, $\sigma_{CE,\ CH3}$= 3.749 Å and $\sigma_{CE,\ CH2}$= 3.972 Å [47]. The effective diameter of any straight-chain alkane can be expressed as follows:

$$\sigma_{\mathrm{CE}} = \left(\frac{2\sigma^3_{CE,CH_3} + (n_c - 2)\sigma^3_{CE,CH_2}}{n_c}\right)^{1/3} \tag{10}$$

Where, σ_{CE} represents the Chapman-Enskog collision diameter of the molecule, Å; n_c represents the carbon number of the alkane.

2.1.3. Capillary entrance effect corrected by slip effect

In viscosity-dominated low-Reynolds number flows, when a liquid enters micro- and nanopores from a larger open space such as an almost velocity-free tank or a larger hydraulic fracture and forms Poiseuille flow, the streamlines of the fluid shrink and bend, and the velocity gradient increases, generating additional flow resistance at the capillary inlet. This additional pressure drop beyond the Poiseuille viscous pressure drop is the entrance effect (also known as Sampson resistance).

Taking into account the slippage of fluid in nanopores, the pressure drop caused by the modified entrance effect can be expressed as [22]

$$R_{entrance}^{CS} = \frac{C_{CS}\mu_{\mathrm{eff,w}}}{2r^3} \tag{11}$$

Where, $R_{entrance}^{CS}$ represents the flow resistance caused by entrance effect, kg/(m·s^2); C_{CS} represents the correction coefficient of the Sampson equation, dimensionless.

Heiranian et al. gave the expression of the correction coefficient [7]

$$C_{CS} = 6\frac{1+2\left(\frac{\alpha}{1+\alpha}\right)^{3/2} - 3\left(\frac{\alpha}{1+\alpha}\right)}{(1+\alpha)^{3/2}} \tag{12}$$

$$\alpha = \frac{L_{s,w}}{\mathrm{r}} \tag{13}$$

Where, α represents the ratio of the water phase slip distance to the pore radius, dimensionless. When $\alpha \rightarrow 0$ (i.e., the slippage distance is small enough or the pore radius is large enough), the equation will degenerate into the classic Sampson equation.

Then the corrected capillary entrance effect pressure drop can be expressed as

$$p_{en} = R_{entrance}^{CS} \times Q = \frac{3\pi\mu_{\mathrm{eff,w}}v}{r}\left[\frac{1+2\left(\frac{\alpha}{1+\alpha}\right)^{3/2} - 3\left(\frac{\alpha}{1+\alpha}\right)}{(1+\alpha)^{3/2}}\right] \tag{14}$$

2.1.4. Gravity

This paper only considers the gravity difference of the saturated crude oil in the original capillary and the imbibition agent, that is, the gravity caused by the difference in oil and water density within the imbibition range.

$$p_g = (\rho_w - \rho_o)gx\sin\beta \tag{15}$$

Where, g represents the acceleration of gravity, m/s^2; β represents the angle between the capillary and the horizontal direction, °.

2.1.5. Three-phase contact line friction and dynamic contact angle

During the spontaneous imbibition process, the liquid front is an oil-water-solid three-phase contact line. The energy dissipation generated by the movement of the contact line causes the liquid to move and spread or shrink on the solid surface, and its contact angle deviates from the static equilibrium value. When the contact line advances (spreads), a larger advancing angle is formed; when the contact line retreats (shrinks), a smaller receding angle is formed. The dynamic wetting angle model based on molecular dynamics theory is [27]

$$\cos\theta_d = \cos\theta_{st} - \frac{\zeta}{\gamma}v \tag{16}$$

$$\zeta = \frac{\left(\mu_{w,i}V_{w,b}\right)^{\omega_w}\left(\mu_{o,i}V_{o,b}\right)^{\omega_o}}{\lambda^3} \tag{17}$$

$$\omega_\mathrm{i} = \mu_i \Big/ \sum_{i=o,w}\mu_i \tag{18}$$

Where, θ_d represents the dynamic contact angle, °; θ_{st} represents the static contact angle, °; γ represents the oil-water IFT, N/m; ζ represents the friction coefficient of the three-phase contact line, Pa·s; V represents the volume of fluid molecules, m^3, $V = 4/3\pi(\sigma_{CE}/2)^3$; λ represents the fluid molecular jump length, which is taken as the effective diameter of the larger molecule in the fluid (i.e., σ_{CE}), m.

The frictional pressure drop generated by the three-phase contact line can be expressed as:

$$p_{fr} = \frac{2\zeta}{r}v \tag{19}$$

2.1.6. Diffusion-osmotic pressure

Diffusion-osmotic pressure refers to the concentration gradient of solutes such as salt ions and surfactant molecules at the oil-water interface, which leads to an osmotic pressure difference along the capillary axis caused by the imbalance of the chemical potential of the solvent on both sides of the interface. This pressure difference points from the high concentration side to the low concentration side (the water phase to the oil phase). If the diffusion-osmotic pressure is of the same magnitude as the capillary pressure, it may change the wetting behavior and even induce interfacial countercurrent or solute-driven spontaneous flow.

The diffusion-osmotic pressure of an ideal dilute solution can be described by the van't Hoff equation [48]

$$\prod = \mathrm{i}R_gTc \tag{20}$$

$$i = 1 + \omega(\mathrm{n} - 1) \tag{21}$$

Where, $\prod$ represents the diffusion-osmotic pressure, Pa; i represents the van't Hoff factor, dimensionless; R_g represents the universal gas constant, 8.31451 m^3·Pa·mol^{-1}·K^{-1}; T represents the absolute temperature, K; c represents the molar concentration of the solution, mol/m^3; ω represents the degree of dissociation, dimensionless; n represents the number of particles after theoretical dissociation, dimensionless.

The osmotic pressure gradient caused by the concentration field along the pore axis can be expressed as

$$\frac{dp_{\prod}}{dx} = \frac{d\prod}{dx} = \mathrm{i}R_gT\frac{dc}{dx} \tag{22}$$

The pressure drop caused by the osmotic pressure difference on both sides of the oil-water interface can be expressed as

$$p_{\prod} = \prod(c) - \prod(0) = \mathrm{i}R_gT\Delta c \tag{23}$$

2.1.7. Capillary pressure

Capillary pressure is the pressure in a capillary that causes a liquid that is wetted or non-wetted with its wall to naturally rise or fall. It is directly related to the IFT, contact angle, and capillary radius, and can reflect the interaction between liquid and liquid, and liquid and solid.

$$p_c = \frac{2\gamma\cos\theta}{\mathrm{r}} \tag{24}$$

During the surfactants imbibition process, the transfer and distribution of surfactants between the water and oil phases has a direct impact on the IFT. This phenomenon of IFT evolving dynamically over time has been widely observed [32]. This research adopt an exponential decay model to describe this dynamic process, as it provides the best fit for the experimental data of the AES surfactant used in this study (with an R^2 of 0.9712, see fitting results in Supplementary Material S.1). Physically, this model reflects the system monotonically approaches equilibrium.

$$\gamma(t) = \gamma_{eq} + \left(\gamma_0 - \gamma_{eq}\right)e^{-k_\gamma t} \tag{25}$$

Where, γ_{eq} represents the equilibrium oil-water IFT, N/m; γ_0 represents the initial oil-water IFT, N/m; k_γ represents the kinetic constant related to the surfactant diffusion/adsorption rate, s^{-1}.

If the IFT decreases too rapidly, strong Marangoni flow may form at the oil-water interface, leading to intensified oil-water mixing and even interface breakdown, forming a microemulsion.

$$p_c = \frac{2\left(\gamma_{eq} + \left(\gamma_0 - \gamma_{eq}\right)e^{-k_\gamma t}\right)\left(\cos\theta_{st} - \frac{\zeta}{\gamma}v\right)}{\mathrm{r}} \tag{26}$$

Substituting the above equation into Eq.1 yields the pore-scale imbibition model:

$$\begin{aligned}\frac{d[\rho_w x v + \rho_o (L-x)v]}{dt} = {} & \frac{2\left(\gamma_{eq} + \left(\gamma_0 - \gamma_{eq}\right)e^{-k_\gamma t}\right)\left(\cos\theta_{st} - \frac{\zeta}{\gamma}v\right)}{\mathrm{r}} - \frac{8\mu_{\mathrm{eff,o}}(L-x)}{(r^2 + 4L_{s,o}r)}\mathrm{v} - \frac{8\mu_{\mathrm{eff,w}}x}{(r^2 + 4L_{s,w}r)}\mathrm{v} \\ & - \frac{3\pi\mu_{\mathrm{eff,w}}v}{r}\left[\frac{1 + 2\left(\frac{\alpha}{1+\alpha}\right)^{3/2} - 3\left(\frac{\alpha}{1+\alpha}\right)}{(1+\alpha)^{3/2}}\right] - (\rho_w - \rho_o)gx\sin(\beta) - \frac{2\zeta}{r}v + \mathrm{i}R_g T\Delta c\end{aligned} \tag{27}$$

Simplifying, we obtain

$$\begin{aligned}\frac{d^2x}{\partial t^2} = {} & \left\{2\left(\gamma_{eq} + \left(\gamma_0 - \gamma_{eq}\right)e^{-k_\gamma t}\right)\left(\cos\theta_{st} - \frac{\zeta}{\gamma}v\right)\Big/ r - 8\mu_{\mathrm{eff,o}}(L-x)\Big/\left(r^2 + 4L_{s,o}r\right)\mathrm{v}\right. \\ & -8\mu_{\mathrm{eff,w}}x\Big/\left(r^2 + 4L_{s,w}r\right)\mathrm{v} - \left(3\pi\mu_{\mathrm{eff,w}}v\Big/r\right)\left[\left(1 + 2\left(\frac{\alpha}{1+\alpha}\right)^{3/2} - 3\left(\frac{\alpha}{1+\alpha}\right)\right)\Big/(1+\alpha)^{3/2}\right] \\ & \left. -(\rho_w - \rho_o)gx\sin(\beta) - \frac{2\zeta}{r}v + \mathrm{i}R_g T\Delta c - \rho_w\left(\frac{dx}{dt}\right)^2 + \rho_o\left(\frac{dx}{dt}\right)^2\right\}\Big/ \rho_w x + \rho_o(L-x)\end{aligned} \tag{28}$$

To determine the flow dynamics, the governing equation (Eq. 28) is solved under defined initial and boundary conditions. Initially ($t = 0$), the nanopore is fully saturated with the oil phase, and the surfactant solution-oil interface is located at the inlet ($x(0) = 0$) with zero initial velocity ($x'(0)= 0$). The calculation proceeds until the interface reaches the pore outlet ($x = L$). Given that the imbibition mechanism evolves from inertial to viscous dominance, the system exhibits significant stiffness. Consequently, this research adopts a variable step and variable order (VSVO) solver based on 1st to 5th order numerical differentiation formulas (NDF) to ensure numerical stability and computational accuracy [49,50].

2.2. Derivation of control equation for nanoporous media imbibition process

Previous studies employed a single uniform capillary tube model to describe the imbibition process, which can effectively reveal the competitive relationship between imbibition pressure and resistance. However, as shown in Fig. 2, the pore structure of natural cores is far more complex than that of ideal capillaries, with pore size distribution showing obvious multi-scale characteristics and flow channels having significant tortuosity in space. This geometric heterogeneity directly affects the imbibition speed, time, and final efficiency of the fluid.

Therefore, this paper introduces the fractal dimension of pore radius and tortuosity to characterize the multi-scale features of pore size distribution and morphology, and establishes a more universal mathematical model for core imbibition.

The actual pores are not ideal straight lines but are tortuous and even branched. The tortuosity fractal dimension D_T is used to describe the variation law of the tortuosity degree of the seepage channels with scale. According to the analysis of Yu [51] et al., in a single tortuous capillary, the climbing length L_t of the capillary liquid column along the actual pore channel and the straight-line projection length L_0 can be expressed as

$$\mathrm{L}_t = \mathrm{L_0}^{D_T} r^{1-D_T} \tag{29}$$

Where, L_t represents the climbing length of the capillary liquid column along the actual pore channel, m; L_0 represents the straight-line projection length, m; D_T represents the tortuosity fractal dimension, dimensionless. For two-dimensional pores, $1 < D_T < 2$, and for three-dimensional pores, $1 < D_T < 3$.

Tang [52] et al. provided the calculation method of D_T

$$D_T = 1 + \frac{\ln \tau_{av}}{\ln \frac{L_0}{r_{av}}} \tag{30}$$

Where, τ_{av} represents the average tortuosity, dimensionless; r_{av}

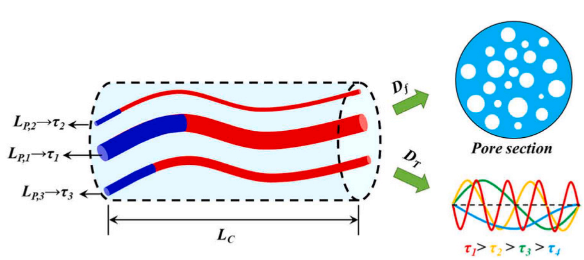

Fig. 2. Fractal characteristics of tortuosity and radius of core pores.

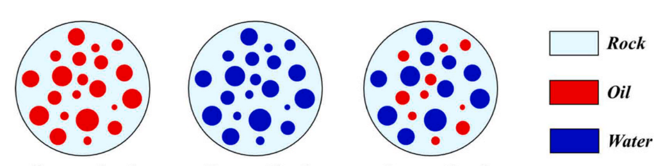

Fig. 3. Calculation method for water saturation along the core.

represents the average pore radius, m.

According to the fractal theory, the average pore radius can be expressed as

$$r_{av} = \frac{D_f}{D_f - 1} r_{\min} \tag{31}$$

Where, D_f represents the pore radius fractal dimension, dimensionless; r_{min} represents the minimum pore radius, m.

Yu [53] et al. assumed that in porous media, some particles are not constrained and overlap each other, while some do not overlap. By taking the average of the tortuosities of these two situations, they obtained the calculation method of the average tortuosity.

$$\tau_{av} = \frac{1}{2}\left[1 + \frac{1}{2}\sqrt{1-\varphi} + \sqrt{1-\varphi}\,\frac{\sqrt{\left(\frac{1}{\sqrt{1-\varphi}} - 1\right)^2 + \frac{1}{4}}}{1 - \sqrt{1-\varphi}}\right] \tag{32}$$

For pore structures that conform to the fractal law, the number of pores with pore diameters greater than r in the rock core has a power function relationship with the pore radius r.

$$\mathrm{N}(\mathrm{R} \geq \mathrm{r}) = \left(\frac{r_{\max}}{r}\right)^{D_f} \tag{33}$$

Then, the number of capillaries in the radius range from r to $r + dr$ can be calculated by the method proposed by Xia [54] et al.

$$-dN = D_f r_{\max}^{D_f} r^{-(D_f+1)} dr \tag{34}$$

The corresponding pore radius probability density distribution of the fractal porous medium is

$$\mathrm{f}(r) = D_f r_{\min}^{D_f} r^{-(D_f+1)} \tag{35}$$

Yu [55] et al. proposed the calculation equation of the pore radius fractal dimension D_f

$$D_f = d_E - \frac{\ln \varphi}{\ln(r_{\min}/r_{\max})} \tag{36}$$

Where, d_E represents the Euclidean space dimension, dimensionless, and for two-dimensional pore cross-sections, $d_E = 2$.

Based on the pore radius probability density function of the fractal porous medium, a group of capillary bundles conforming to the corresponding distribution was generated using Matlab, and the oil-water distribution on different cross-sections of the rock core was obtained according to the imbibition model and the tortuosity calculation method established in the previous text (Fig. 3). The water saturation of the cross-section can be calculated by the Eq.37.

$$S_w = \frac{\sum_{Pore_w} A_j}{\sum_{Pore_{o,w}} A_i} \tag{37}$$

Where, S_w represents the water saturation, dimensionless; A represents the pore cross-sectional area, m^2.

3. Results and discussion

3.1. Model validation

Wang [56] et al. conducted numerical simulations of the oil-water imbibition process in nanopores based on the LBM method, considering the effects of nanoscale effects, dynamic contact angles, and entrance effects on the imbibition process. The results obtained were relatively reliable. This research used these data as a basis to validate the model, and the parameters required for validation can be referred to in that article. As shown in Fig. 4a, the model proposed in this paper is relatively close to the results in the literature, especially in the early stage of imbibition with the final error 6.1 %. The results indicate that in nanopores, the viscous adsorbed layer effect, entrance effect, and slip effect have a significant impact on the imbibition process. However, the dynamic contact angle has a relatively small effect on the imbibition effect. This is because the movement speed of the oil-water front in 5 nm pores is small, making it difficult to generate a strong dynamic wetting effect. In larger pores, the dynamic contact angle may not be negligible. Therefore, subsequent studies did not ignore any content in the model. Similarly, the imbibition distances calculated by the model proposed in this paper for capillaries with different radii also exhibit good consistency with the results from the reference (Fig. 4b).

3.2. Results of single nano capillary imbibition

This section conducts research on the imbibition process of oil-surfactant in a single capillary based on the established model. The parameters involved in the model are shown in Table 1, where the parameters listed are the basic ones. Notably, the values selected for the model are derived from experiments or literature, with most being common values in oil and gas field development (for example, a static contact angle θ_{st} of 30° is selected to represent a typical water-wet reservoir condition). When the research factors change, the variable values will change accordingly, and the rest of the factors will take the basic values.

3.2.1. Analysis of the influence of different parameters on imbibition

The capillary radius primarily influences the P_c, P_{en}, P_v and P_{fr} during

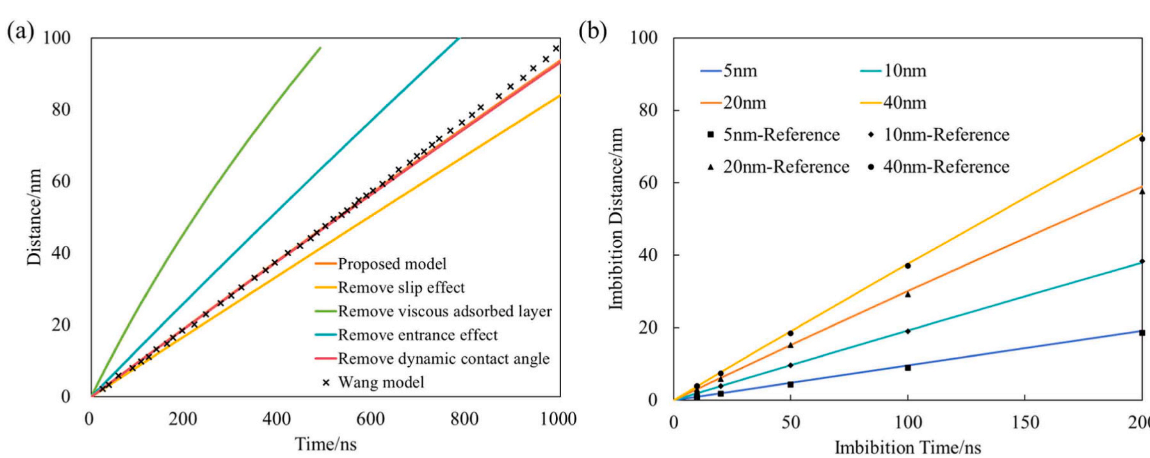

Fig. 4. Comparison of pore imbibition model and reference result [56].

Table 1
The parameter values involved in the imbibition model of this research.

Parameter	Value	Source	Parameter	Value	Source
ρ_o, kg/m^3	850	Experimental measurement	r, m	5×10^{-9}	Model setting
ρ_w, kg/m^3	1000	Standard value	L, m	5×10^{-2}	Model setting
n_c, dimensionless	10	Model setting	μ_w, Pa·s	0.001	Reference [56]
g, m/s^2	9.8	Standard value	μ_o, Pa·s	0.0076	Experimental measurement
T, K	293	Model setting	θ_{st}, °	30	Reference [56]
R_g, m^3·Pa/mol/K	8.31451	Standard value	c, mol/m^3	1.3719	Model setting
i, dimensionless	2	Supplementary Material S.1	γ_{eq}, N/m	0.0011062	Fitted from exp. data [32]
γ_0, N/m	0.01773	Fitted from exp. data [32]	k_γ, s^{-1}	0.00448	Fitted from exp. data [32]
β, °	30	Reference [56]	$\sigma_{CE,\ water}$, Å	2.641	Reference [46]
$\sigma_{CE,\ CH2}$, Å	3.972	Reference [47]	$\sigma_{CE,\ CH3}$, Å	3.749	Reference [47]

the imbibition process. As shown in Fig. 5a and b, with the increase of capillary radius, the imbibition distance and imbibition velocity increase. At the same time (T = 180,000 s), the imbibition distance of 1 nm pores is only 0.0037 m, while that of 100 nm pores can reach 0.05 m, which is 13.5 times that of 1 nm pores. During the extended plateau period (indicated by black circles in Fig. 5b), the imbibition velocities for 3 nm, 5 nm, 10 nm, 50 nm, and 100 nm pores are 1.5, 1.9, 2.4, 4.5, and 7.1 times that of 1 nm pores, respectively. This occurs because the primary driving pressure for imbibition, capillary pressure ($2\pi r\gamma cos\theta$), increases with radius. Additionally, the viscous adsorbed layer effect and viscous pressure drop is more prominent in smaller pores.

The viscosity of the oil phase mainly affects the P_{en}, P_v and P_{fr} during the imbibition process. As shown in Fig. 5c and d, as the viscosity of the oil phase increases from 1 mPa·s to 100 mPa·s, the imbibition distance increases by 4.6 times at the same time (T = 1200,000 s), and the time required to reach the same imbibition distance (0.01 m) increases by 12.85 times.

The viscosity of the oil phase increases the flow resistance of the imbibition process by affecting the viscous pressure drop of the oil phase, significantly reducing the imbibition velocity and limiting the imbibition distance. However, at the final moment, the imbibition velocities of different oil phase viscosities are close. This is because at this time, the wetting phase has basically completely replaced the non-wetting phase through imbibition. According to Eqs. 4 and 6, the imbibition velocity is independent of the viscosity of the oil phase at this time.

The IFT and contact angle mainly affect the P_c during the imbibition process. As shown in Fig. 5e and f, under the condition of IFT < 0.1 mN/m, the imbibition process hardly occurs. From 0.001 mN/m to 10 mN/m, the imbibition velocity increases by 5 orders of magnitude. Similarly, in the non-wetting condition ($\theta \geq 90°$), the imbibition process only occurs a small amount of displacement under the influence of diffusion-osmotic pressure. A strong wetting environment is conducive to the imbibition process (Fig. 5g and h). From the perspective of imbibition kinetics, increasing the IFT and improving wettability are beneficial to the imbibition process.

The diffusion-osmotic pressure of surfactant mainly affects the $P_{\prod}$ during the imbibition process. As shown in Fig. 5i and j, a higher diffusion-osmotic pressure (> 67kPa) of surfactant has a significant impact on the imbibition process. The time required for a 668kPa diffusion-osmotic pressure surfactant to reach the same imbibition distance (0.02 m) is only half that of a 1 % one. The diffusion-osmotic pressure of low-concentration surfactants has little effect on the imbibition process. As another important driving pressure of the imbibition process, diffusion-osmotic pressure directly affects the imbibition velocity and should not be ignored in the imbibition process of nanopores.

To ensure the practical applicability of the model, the parameter ranges in this research were selected to represent typical conditions of tight reservoirs and surfactant (e.g., pore radii of 1–100 nm and IFT of 0.001–10 mN/m). To quantitatively evaluate the influence of each parameter, the Pearson correlation coefficient R was calculated. The results indicate that IFT (R=0.9996) and diffusion-osmotic pressure (R=0.9995) are the most significant controlling factors, followed by contact angle (R=0.9811), pore radius (R=0.9710), and oil-phase viscosity (R=0.6891).

3.2.2. Dynamic characteristics of nanopore imbibition

During the imbibition process, the shape of the time-distance curve is determined by the competition and coupling of various mechanical factors, presenting three types: convex (Type I), linear (Type II), and concave (Type III) (Fig. 6a). The ratio of the bulk fluid in the pores to the total pore cross-sectional area is defined as ε(Fig. 6a). When the capillary radius is large, the mass of the liquid column and the slip effect cannot be ignored, the fluid flow rate is fast, and the inertial effect is significant (Fig. 6b), while the viscous resistance is small, resulting in a convex overall shape of the imbibition curve (Type I). In the case of extremely small radius, the interaction between liquid molecules and solid walls is strong, viscous resistance become dominant, and the adsorption layer effect at the interface is enhanced, while the inertial effect rapidly decays, causing the curve to be concave (Type III) (Fig. 6a, c). Additionally, factors such as pressure drop at the inlet, the diffusion-osmotic pressure of the solution, and gravity will further modify the balance between driving pressure and resistant pressure under different scales and conditions, resulting in different curve shapes.

Capillary pressure and diffusion-osmotic pressure are the driving pressures of the imbibition process, while inertial pressure, viscous pressure drop, entrance effect pressure, gravity, and frictional pressure are the resistances. Under the combined action of driving pressure and resistance, the imbibition process exhibits complex characteristics (Fig. 6b, d). Based on the time-velocity curves of capillaries with different radii, the imbibition process can be divided into five stages:

The first stage (S1) Inertial Initiation Stage: At the initial stage of imbibition (T < 1.5 ns), capillary pressure and inertial pressure jointly drive the fluid to accelerate into the pore, and since the inertial effect has not yet been balanced by viscosity (highlighted by the red circle in Fig. 6d, which indicates the inertial-dominated initiation zone), the flow rate shows a rapid upward trend, demonstrating typical inertial initiation characteristics;

The second stage (S2) Quasi-Steady Imbibition Stage: As the flow develops (T < 10 s), the inertial effect persists, and with the increase in flow rate, the imbibition resistance increases (Fig. 6b, d), the system enters a quasi-steady state where the driving pressure and resistance are balanced, and the imbibition velocity remains constant. At this time, the fluid has not yet entered the pore in large quantities (Fig. 6a), and the imbibition distance is less than 3.77×10^{-6} m;

The third stage (S3) Inertial-Adhesive-Frictional Decay Stage: As time increases (T < 1000 s), the inertial effect weakens, and the decrease in flow rate leads to a decline in the resistance related to the imbibition velocity. However, as the fluid begins to enter the capillary, the viscous pressure drop of the water phase fluctuates due to the dual influence of the liquid column length and imbibition velocity. This fluctuation is explicitly marked by the blue circles in Fig. 6c and d, representing the instability in aqueous phase viscous pressure. Although

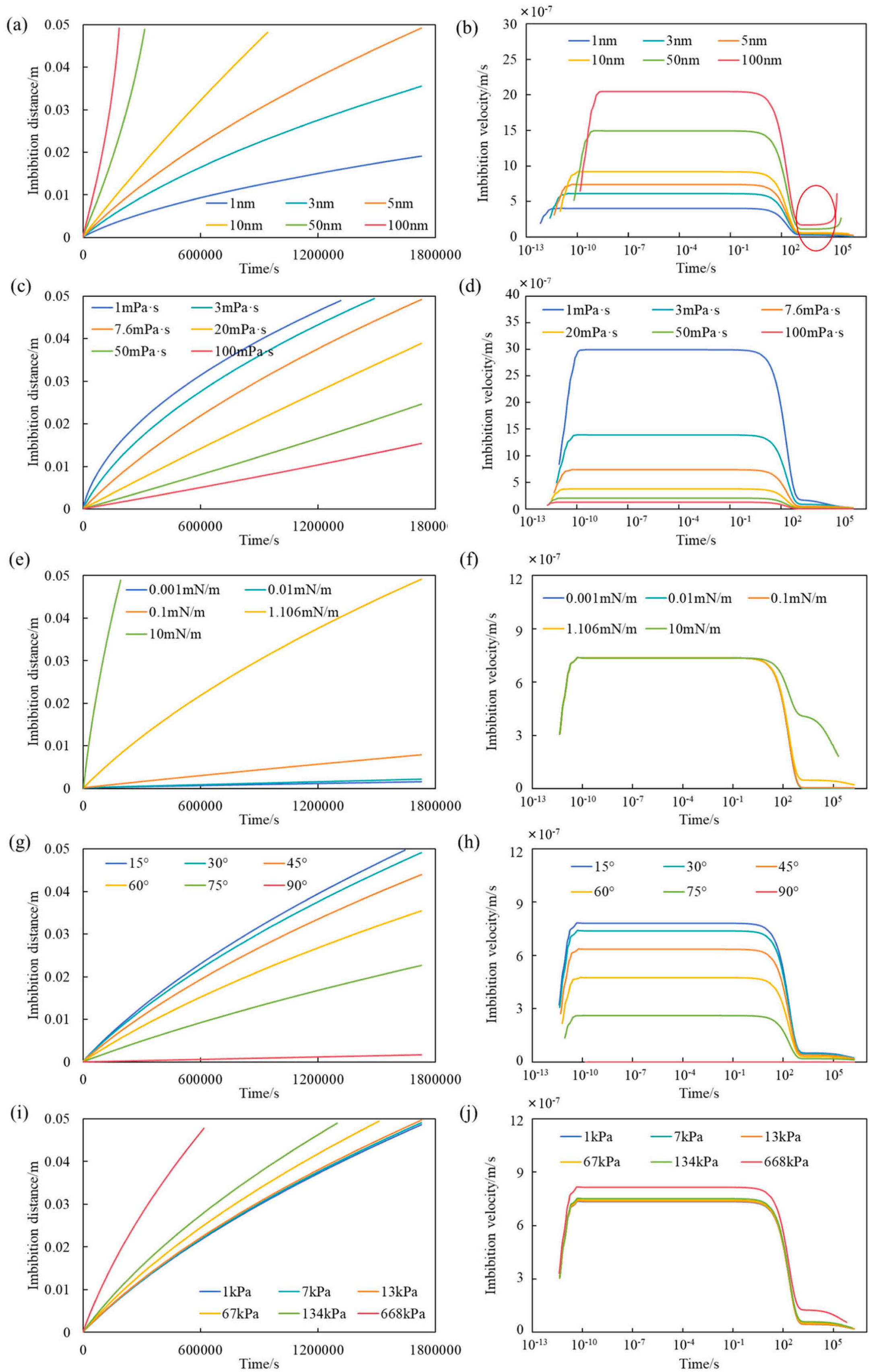

Fig. 5. Impact of key parameters on spontaneous imbibition kinetics in single nanopores. (a, b) Pore radius effects: Imbibition distance (a) and velocity (b) versus time. (c, d) Oil viscosity effects: Imbibition distance (c) and velocity (d) versus time. (e, f) Interfacial tension (IFT) effects: Imbibition distance (e) and velocity (f) versus time. (g, h) Contact angle effects: Imbibition distance (g) and velocity (h) versus time. (i, j) Diffusion-osmotic pressure effects: Imbibition distance (i) and velocity (j) versus time.

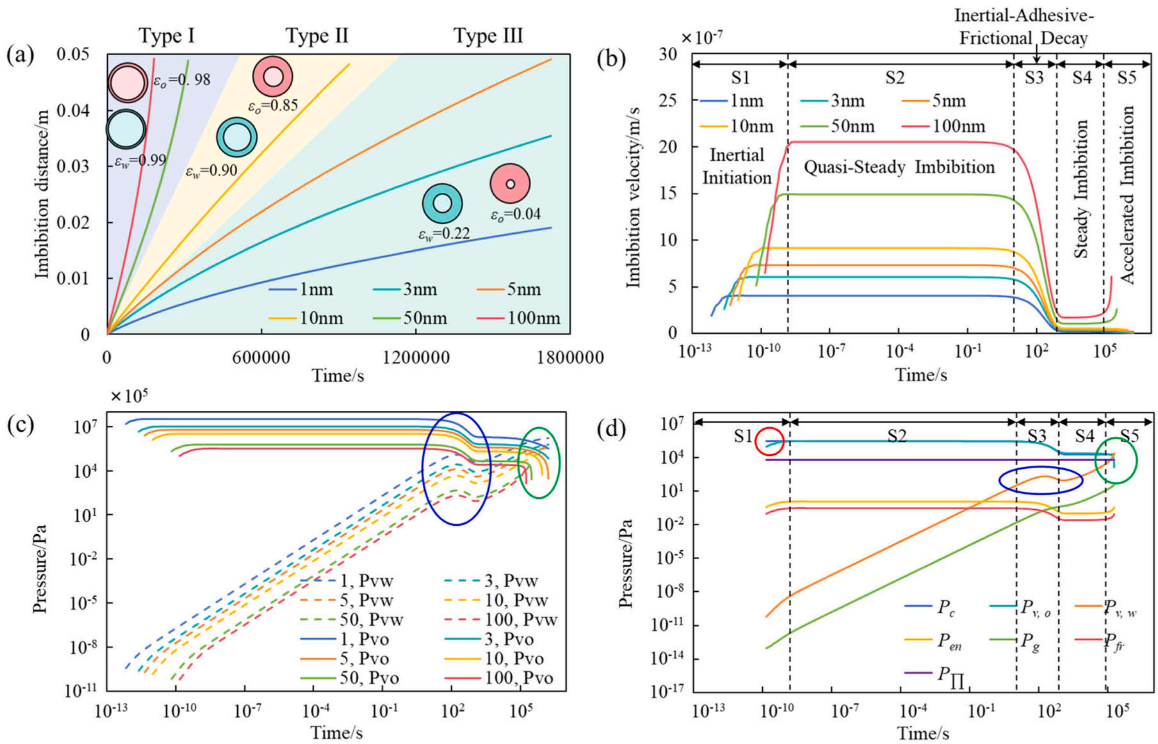

Fig. 6. Evolution of mechanical mechanisms and flow stages during spontaneous imbibition. (a) Imbibition distance-time curves for varying pore radii, identifying three curve types (Type I, II, III). (b) Imbibition velocity-time curves illustrating the five flow stages (S1–S5). (c) Comparison of viscous pressure drops between oil and water phases over time. (d) Dynamic competition of driving and resisting pressures for a 5 nm nanopore.

fluctuating, the viscous pressure drop of the water phase still differs by orders of magnitude from that of the oil phase, and the viscous pressure drop of the oil phase dominates the flow characteristics of this process.

The fourth stage (S4) - Steady Imbibition Stage: This stage lasts for a relatively long time ($T < 10^5$s), similar to S2, but the inertial effect is almost non-existent, resulting in a true balance between the driving and resisting pressures of the imbibition process. The imbibition rate remains at a relatively low level (Fig. 6b).

The fifth stage (S5) - Accelerated Imbibition Stage: As the water phase continues to enter the capillary, the viscous pressure drop of the water phase gradually increases to exceed that of the oil phase (indicated by the green circles in Fig. 6c and d, which mark the pressure crossover point). Due to the lower viscosity of the water phase, the overall viscous pressure drop within the capillary decreases, and the imbibition rate begins to increase. However, this phenomenon only occurs in larger capillaries ($r > 10$ nm), which is the reason for the formation of the convex imbibition curve. This indicates that the prerequisite for this phenomenon is that the water phase can enter most of the capillaries, and the viscous pressure drop of the water phase can exceed that of the oil phase at an earlier time. In smaller pores, the viscous adsorbed layer effect is significant, and the distance over which the water phase acts is limited within the set end time, and the phenomenon of an increase in the final imbibition rate does not occur.

3.3. Results of nanoporous media imbibition

The parameters for oil-water-rock interface properties in this section are shown in Table 1. Additionally, this study defines a physical factor χ to characterize the variation in pore distribution characteristics of the core, which is expressed as the ratio of the maximum (minimum) pore radius of two cores. For more details, please refer to Supplementary Material S. 2.

3.3.1. Analysis of the influence of different parameters on imbibition efficiency and velocity

As shown in Fig. 7, the core imbibition efficiency-time relationship under different conditions generally conforms to a square root relationship. In most cases, the imbibition efficiency of the 5 cm tight core is generally less than 20 %, and the imbibition distance is generally less than 1 cm when the water saturation is 0.2. The pores in the rock core are longer since the high tortuosity, and the viscous pressure drop during the imbibition process is greater, so the imbibition efficiency and imbibition distance may be lower. Therefore, hydraulic fracturing is an important approach for developing such reservoirs. The water saturation-distance curve usually shows a trend of rapid decline followed by a slow decline, with a distinct inflection point. It is not difficult to find that the distance of the inflection point reflects the sweep distance of most pores in the core. To analyze the characteristics of the imbibition process, this study defines the full sweep distance Lfs and the water saturation at the outlet end Swol (Fig. 8a).

Core pore radius is important factor affecting imbibition efficiency. As χ increases from 0.1 to 10, the imbibition efficiency increases by 10 times, and the initial velocity increases by 1.7 orders of magnitude (Fig. 7a, f). The imbibition process of the large-pore tight core makes the main contribution. When $\chi = 0.1$, the surfactant cannot achieve full sweep in the core (Fig. 8a). With the increase of χ, Lfs and Swol show significant increases (Fig. 8f).

The viscosity of the oil phase also has a significant impact. The imbibition efficiency of crude oil with a viscosity of 100 mPa·s is only 3.3 %, and the velocity is reduced by nearly two orders of magnitude compared to crude oil with a viscosity of 1 mPa·s (Fig. 7b, g). The inflection point distance of high-viscosity crude oil is only 1/10 of that of low-viscosity crude oil, and small pores basically cannot undergo imbibition, resulting in a very small Lfs (Fig. 8b, g). For tight reservoirs, the resistance of the imbibition process of high-viscosity crude oil is too large, and it is necessary to enhance the imbibition driving pressure by

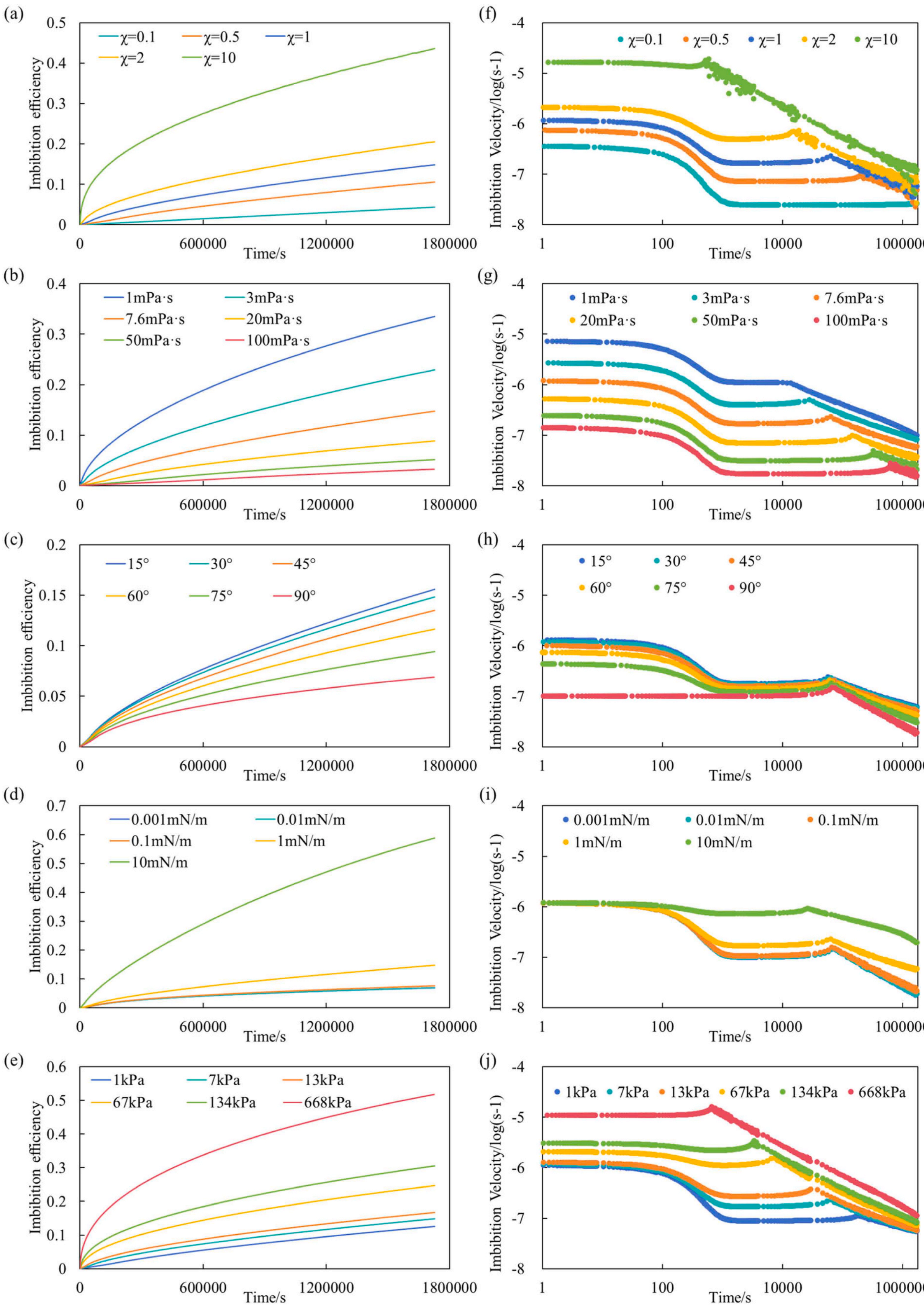

Fig. 7. Core-scale imbibition efficiency and velocity under varying conditions. (a–e) Imbibition efficiency: Time-dependent curves for varying pore structure factor χ (a), oil viscosity (b), contact angle (c), IFT (d), and diffusion-osmotic pressure (e). (f–j) Imbibition velocity: Corresponding logarithmic velocity curves for the same parameter sets (f–j).

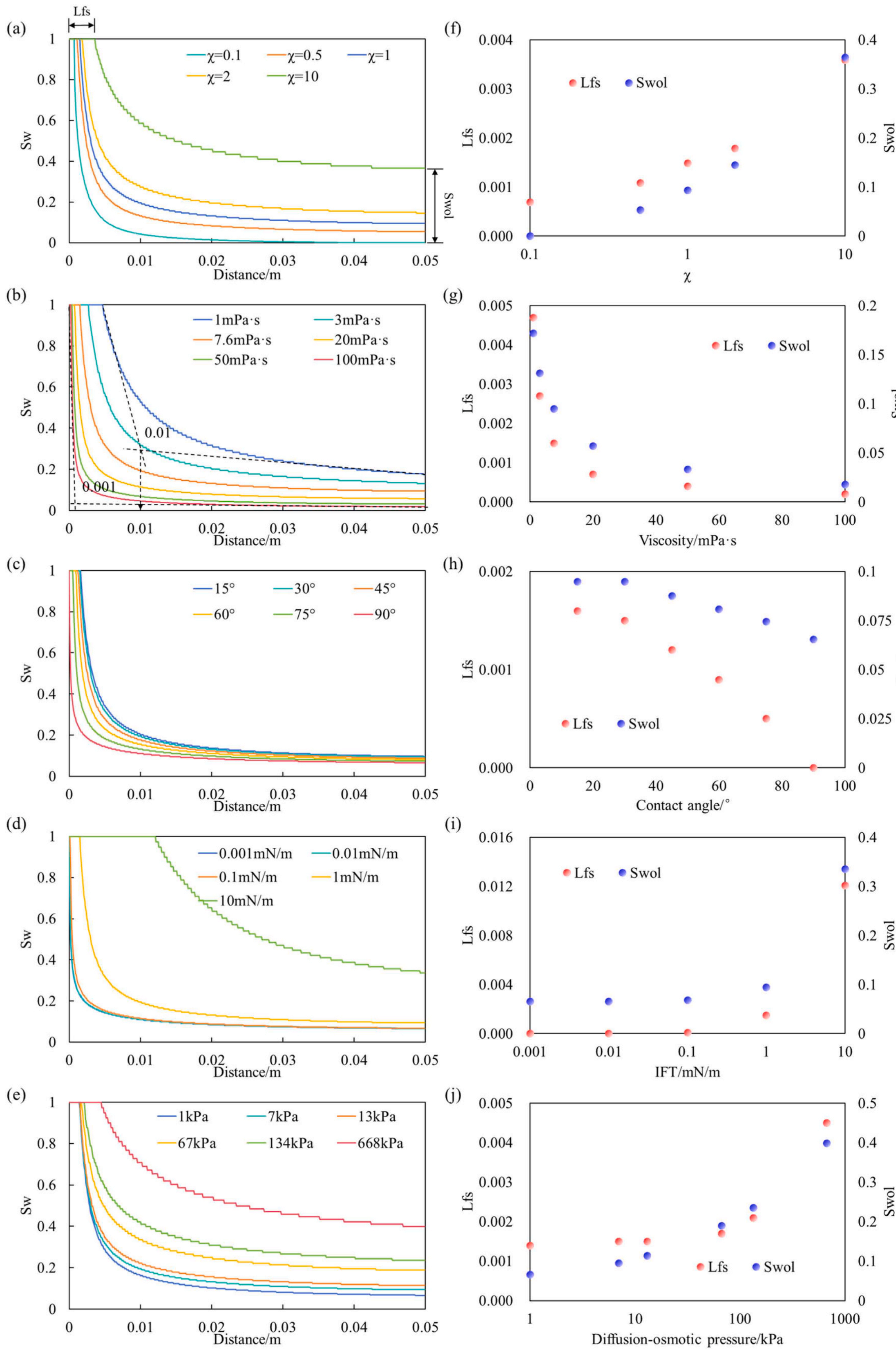

Fig. 8. Spatial distribution of water saturation and sweep characteristics. (a–e) Saturation Sw distribution along the core distance for varying χ (a), viscosity (b), contact angle (c), IFT (d), and diffusion-osmotic pressure (e). (f–j) Full sweep distance Lfs and outlet water saturation Swol for the corresponding parameters.

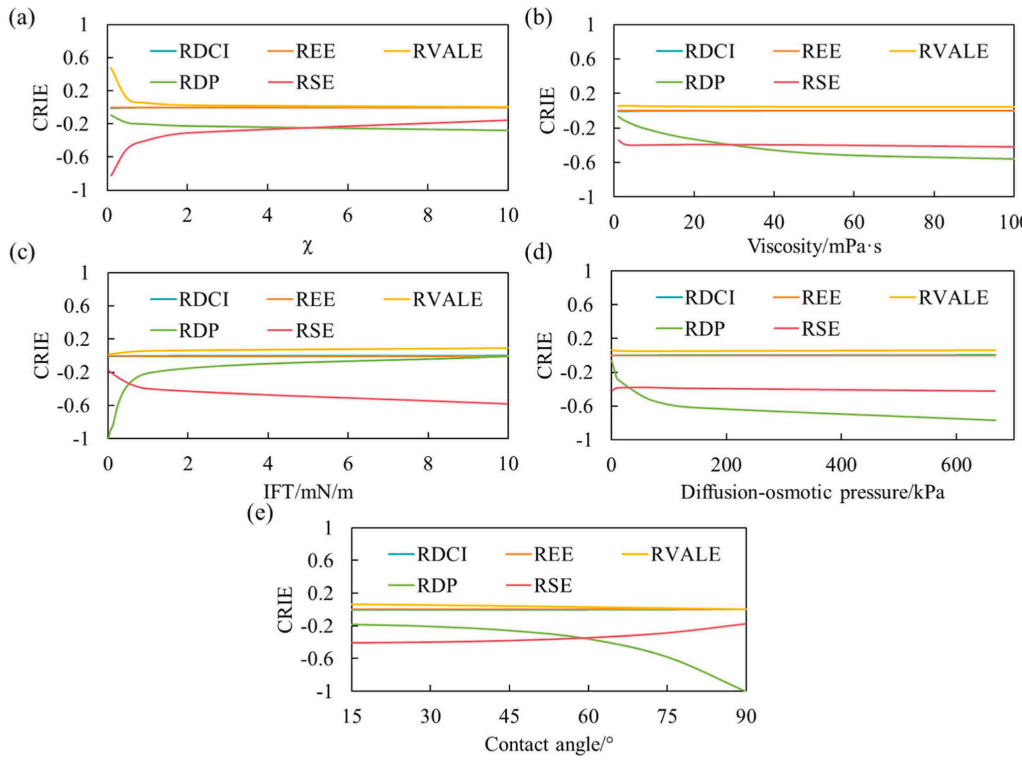

Fig. 9. Differences in imbibition efficiency between different imbibition models and proposed models. (CRIE: Change rate of imbibition efficiency; RDCI: Model remove dynamic contact angle and IFT; REE: Model remove entrance effect; RVALE: Model remove viscous adsorbed layer effect; RDP: Model remove diffusion-osmotic pressure; RSE: Model remove slip effect).

improving wettability or increasing IFT or reduce the imbibition resistance by heating or injecting viscosity reducers. The recommended crude oil viscosity should be less than 10 mPa·s.

Contact angle and IFT affect imbibition efficiency by influencing capillary pressure. A water-wet environment is a prerequisite. As wettability evolves towards hydrophilicity, imbibition efficiency increases, but the cost of further reducing the contact angle when it is less than 30° will significantly increase, while the increase in imbibition efficiency is limited (Fig. 7c, h, Fig. 8c, h). At the same time, as IFT increases, imbibition efficiency significantly increases. When the IFT is less than 0.1 mN/m, capillary pressure has a small impact, and the imbibition efficiency at this time is mainly contributed by diffusion-osmotic pressure (Fig. 7d, i). As shown in Fig. 8d, i, high IFT is of great significance for the development of tight reservoirs. When the IFT reaches 10 mN/m, Lfs and Swol increase significantly (Lfs and Swol are 8 times and 3.3 times that under the condition of 1 mN/m, respectively), and more than 1/5 of the pores can achieve full sweep. Surfactants can reduce IFT, form emulsions, and improve wettability, thereby enhancing the flowability of crude oil. However, the addition of surfactants greatly reduces the driving pressure of imbibition. Therefore, it is necessary to maintain a certain IFT in the reservoir development.

Surprisingly, the effect of diffusion-osmotic pressure on imbibition efficiency is also significant. The high diffusion-osmotic pressure surfactants can increase the imbibition efficiency by 39 % and the imbibition velocity by two orders of magnitude (Fig. 7e, j). Especially when the diffusion-osmotic pressure is greater than 13kPa, both the imbibition efficiency and sweep range are significantly improved (Fig. 8e, j). This indicates that the diffusion-osmotic pressure is not negligible for field development, and the existing methods may underestimate the imbibition efficiency of tight reservoirs under high diffusion-osmotic pressure ionic conditions.

The water saturation-distance chart under different conditions be found in Supplementary Material S. 3.

3.3.2. Influences of different flow effect on imbibition

As shown in Fig. 9, the viscous adsorbed layer effect increases the effective viscosity of the fluid, which is unfavorable. The slip effect increases the fluid velocity, and the diffusion-osmotic pressure enhances the driving pressure, which is beneficial. Due to the overall lower flow velocity in nano-pores and the relatively long overall pore length within the core (up to 0.35 m in actual length), the entrance effect and dynamic contact angle and IFT have a relatively small impact. However, for shorter single capillaries, these effects cannot be ignored (Fig. 4). As shown in Fig. 9a, the viscous adsorbed layer effect has a greater influence in smaller pores, and its effect almost disappears when $\chi > 1$. Therefore, for shale or tight reservoirs with poor properties, the viscous adsorbed layer effect must be considered; otherwise, the productivity of the reservoir will be overestimated. Meanwhile, as the pore radius increases, the imbibition process gradually shifts from being dominated by diffusion-osmotic pressure to being dominated by the slip effect (Fig. 9a). With the decrease in IFT, the increase in crude oil viscosity, diffusion-osmotic pressure, and contact angle enhances the effect of diffusion-osmotic pressure and weakens the effect of the slip effect (Fig. 9b-e). In summary, under conditions of sufficient imbibition driving pressure (low crude oil viscosity, high IFT, and strong hydrophilicity), the imbibition rate is fast, and the slip effect dominates the occurrence of imbibition; under conditions of insufficient imbibition driving pressure, the diffusion-osmotic pressure plays an important role as an additional driving pressure. Consequently, these findings offer distinct strategies for reservoir development. In high-energy scenarios (e.g., near-wellbore or high-permeability zones), maximizing hydraulic pressure can leverage the slip effect to accelerate recovery. In contrast, for ultra-tight reservoirs under low-energy conditions, where hydraulic driving is limited, engineering efforts should focus on optimizing surfactant formulations to establish chemical potential gradients, thereby activating diffusion-osmotic pressure to mobilize trapped oil.

3.4. Practical implications, limitations, and future directions

The proposed kinetic model offers theoretical guidance for optimizing field applications in tight oil reservoirs by clarifying the dominant mechanisms under different flow regimes.

3.4.1. Guidance for field applications

The research reveals a regime transition where diffusion-osmotic pressure becomes the primary driving force under low-energy conditions. This suggests that for ultra-tight formations, engineering strategies should shift from relying on hydraulic pressure to establishing chemical potential gradients. Consequently, optimizing diffusion-osmotic pressure (surfactant concentration) becomes critical; a higher concentration gradient can activate diffusion-osmosis to mobilize oil trapped in nanopores that are inaccessible to conventional pressure driving. Based on the sensitivity analysis, the most effective parameters adjustable through field measures are diffusion-osmotic pressure (surfactant concentration), IFT, and fluid viscosity. Specifically, concentration can be directly regulated during injection preparation to enhance osmotic drive; IFT can be tailored by screening surfactant formulations to improve capillary number; and viscosity can be modified via viscosity reducers or thermal methods to decrease flow resistance.

3.4.2. Limitations and future work

It should be noted that the current model simplifies the porous media using fractal theory and assumes 1D flow, which may not fully capture the complex 3D heterogeneity and cross-flow behaviors in actual reservoirs. Additionally, chemical reactions between surfactants and rock minerals are not explicitly considered. Future work will focus on extending this kinetic framework to 3D pore network modeling and coupling it with reservoir numerical simulators to evaluate macroscopic recovery performance under complex geological conditions.

4. Conclusion

In this research, a comprehensive kinetic model for spontaneous surfactant solution imbibition in nanopores was established. This model systematically integrates dynamic contact angle, dynamic interfacial tension (IFT), viscous adsorbed layer, slip effect, entrance effect and diffusion-osmotic pressure. By incorporating fractal theory, the model successfully scales up from single nanopores to the core scale, enabling quantitative evaluation of imbibition efficiency and sweep distance in tight reservoirs. The main conclusions are as follows:

(1) The results show that the key factors affecting nanoscale pore imbibition, in descending order of significance, are IFT, diffusion-osmotic pressure, contact angle, pore radius, and oil-phase viscosity.
(2) The imbibition dynamics are controlled by the competition between driving and resisting pressures. Accordingly, the process is categorized into five distinct stages: inertial initiation, quasi-steady imbibition, inertial-adhesive-frictional decay, steady imbibition, and accelerated imbibition.
(3) Small pores and high tortuosity limit liquid propagation, while a higher pore structure factor χ significantly improves imbibition efficiency. High oil viscosity significantly increases flow resistance, reducing the sweep distance by 10 times. Contact angle and IFT influence the driving pressure by affecting capillary pressure. Consequently, to maximize capillary driving forces in reservoir development, it is essential to optimize IFT and improve wettability alteration.
(4) The viscous adsorbed layer effect increases the effective fluid viscosity, hindering imbibition in nanopores. The flow mechanisms exhibit a regime transition: the slip effect dominates the flow enhancement under high-energy conditions, whereas diffusion-osmotic pressure becomes the primary driving mechanism under low-energy conditions.

CRediT authorship contribution statement

Yanpei Li: Visualization, Investigation. **Yaqian Zhang:** Visualization, Investigation, Conceptualization. **Zheyu Liu:** Supervision, Resources, Methodology, Funding acquisition. **Tao Song:** Validation, Investigation. **Qihang Li:** Visualization, Validation. **Yiqiang Li:** Supervision, Project administration, Funding acquisition. **Jinxin Cao:** Writing – original draft, Validation, Methodology, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors thank the National Science and Technology Major Project (Grant No. 2025ZD1406206) and the National Natural Science Foundation of China (Grant No. U23B6003) for the support of this work.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.colsurfa.2026.139891.

Data availability

Data will be made available on request.

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